Riding Wavelets: Reproducing “Riding Wavelets” on Open Data

Mini-project (ML for Time Series)

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MVA — Jan 2026

Why this project?

- We do not work in finance: we were curious about **very noisy** data and **non-trivial spectral analysis** around extreme events.
- We reproduced a hedge-fund research preprint: **Aubrun et al., "Riding Wavelets"** (jump classes via wavelets + Kernel PCA).
- Key motivation: markets are **non-stationary** (volatility clustering, intraday patterns) so global templates (e.g., dictionary learning) can be brittle.
- Goal: implement the full pipeline from the paper with **open data**, end-to-end, and test robustness across **time scales** and **markets**.

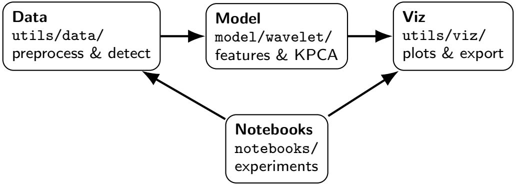
Reproducibility and engineering

- We worked 80+ hours.
- We wrote 2k LOC.
- We made 73 commits.
- We open-source the code.
- We install in one line.

1-liner installation

```
chmod +x launch.sh && ./launch.sh
```

Code structure (high-level)



- We re-implement the full pipeline.

Data and preprocessing

- We use 5GB Stooq OHLC equities.
- We cover 5-min, hourly, daily.
- We study Poland, Hungary.
- We test HK, US, UK.
- We normalize by local volatility.

Jump detection and non-stationarity

- We standardize returns.

$$x(t) = \frac{r(t)}{f(t)\sigma(t)}.$$

- Markets are highly non-stationary.
- We avoid dictionary learning.
- We threshold with $\theta = 4$ (5-min).
- We assume max-stable (Gumbel).

Seasonality $f(t)$ and volatility $\sigma(t)$

- Returns: $r_t = \log p_t - \log p_{t-\Delta}$.
- Intraday bin: $\tau(t) =$ time-of-day of t.
- Robust seasonality estimate:

$$m(\tau) = \mathrm{median}\{|r_t| : \tau(t) = \tau\}, \qquad q = \sqrt{\frac{1}{|\mathcal{T}|}\sum_{\tau\in\mathcal{T}} m(\tau)^2}, \qquad f_t = \frac{m(\tau(t))}{q}.$$

- Deseasonalize: $r_t^{\mathrm{des}} = r_t/f_t$ (daily: $f_t \equiv 1$).
- Local volatility (EWMA std on r^{des}):

$$\mu_t = \alpha r_t^{\mathrm{des}} + (1-\alpha)\mu_{t-1}, \qquad v_t = \alpha\left(r_t^{\mathrm{des}} - \mu_t\right)^2 + (1-\alpha)v_{t-1}, \qquad \sigma_t = \sqrt{v_t}, \qquad \alpha =$$

Wavelet features and interpretable directions

- We align jump signs ($t = 0$).
- We compute complex wavelet scattering.

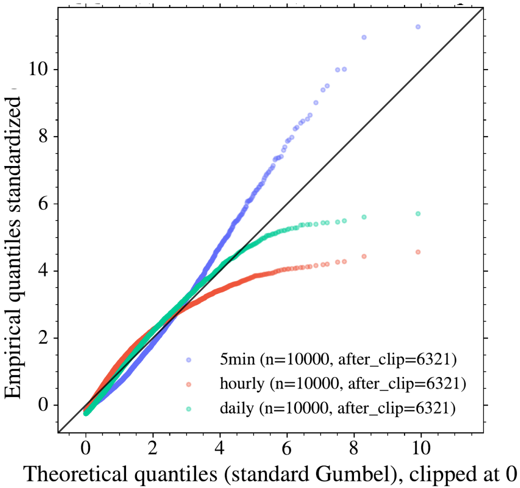
$$\Phi(x) = \big(W_j \bar{x}(0),\ W_{j_2}\,|W_{j_1} x|(0)\big)_{1 \le j \le J,\ 1 \le j_1 < j_2 \le J}.$$

- We run Kernel PCA on $K(\Phi(x_i), \Phi(x_j))$.
- We interpret D_1 as reflexivity.
- We score mean-reversion D_2.
- We score trend D_3.

$$D_2 = x(-1) - x(+1), \qquad D_3 = x(-1) + x(+1).$$

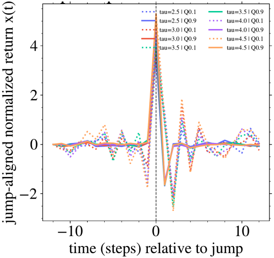
Gumbel diagnostic

Hypothesis: Jump scores are approximately max-stable (Gumbel-like) after normalization.
Plot: Q–Q of $|x(t)|$ vs fitted Gumbel.
What we see: Overall good fit; mild 5-min tail deviation.

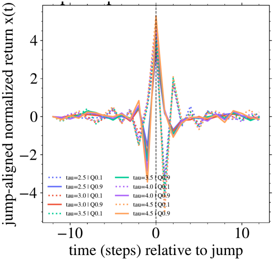


Research question 1: robustness to the threshold (5-min)

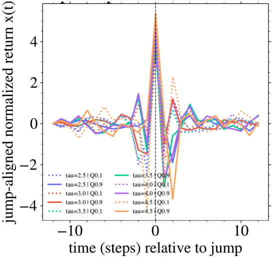
Hypothesis: Average jump profiles stay stable across θ.
Plot: Overlay mean profiles across θ (5-min).
What we see: D_1 and D_3 stay stable; D_2 gets noisier at extremes.



D_1 overlays



D_2 overlays



D_3 overlays

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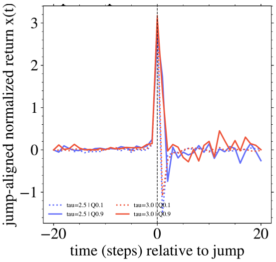
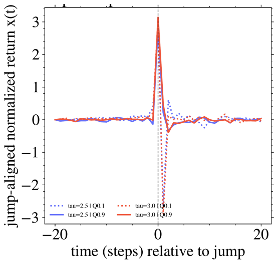
Research question 1 (cont.): daily frequency

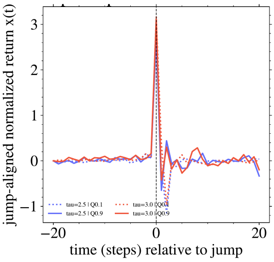
Hypothesis: Same structure holds at daily scale.
Plot: Daily overlays across θ.
What we see: Profiles remain similar, with higher noise.



D_1 (daily)



D_2 (daily)



D_3 (daily)

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Research question 2: do Scattering Spectra help?

Hypothesis: Scattering Spectra adds little at our resolution.
Plot: With-SS vs no-SS ablation (runs).
What we see: Embeddings stay very similar; D_1 correlation stays high.

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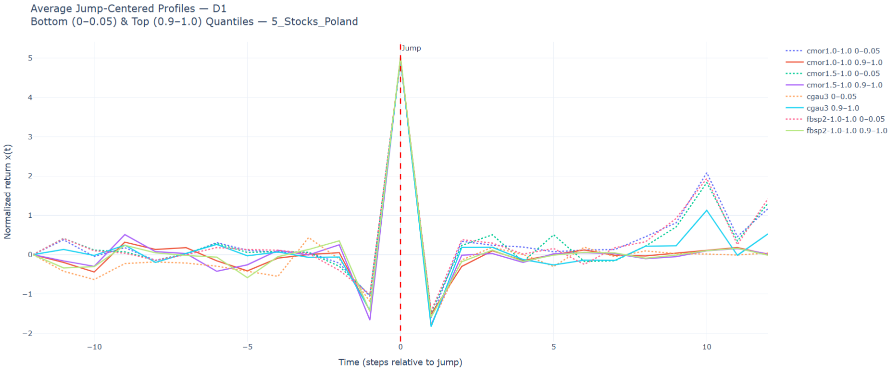
Wavelet robustness: D_1

Hypothesis: D_1 is robust to wavelet choice.
Plot: Different wavelet families (fbsp/cmor/cgau).
What we see: D_1 stays stable across wavelets.



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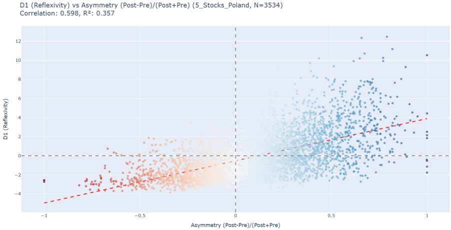
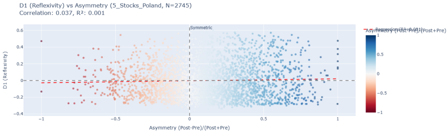
Asymmetry: daily vs 5-min

Hypothesis: Daily asymmetry correlates; 5-min does not.
Plot: Asymmetry correlation diagnostics (daily vs 5-min).
What we see: We see strong daily correlation. We see weak 5-min correlation.



daily



5-min

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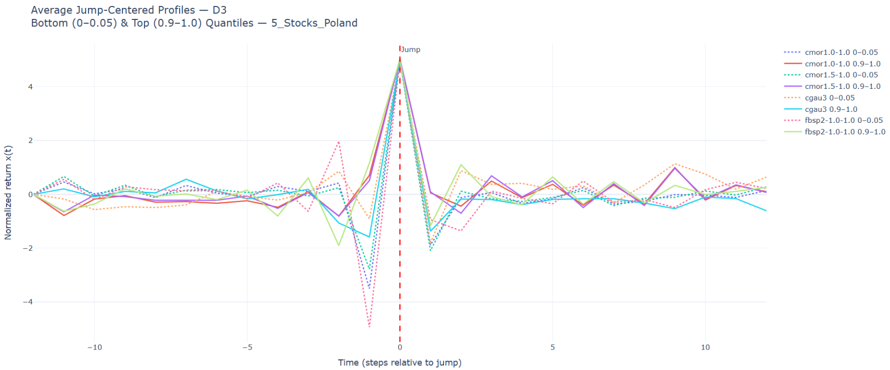
Trend direction: D_3

Hypothesis: D_3 is a stable trend direction.
Plot: D_3 profiles across wavelets / markets.
What we see: D_3 is unstable on PL/HU; we need more data (US placeholder).



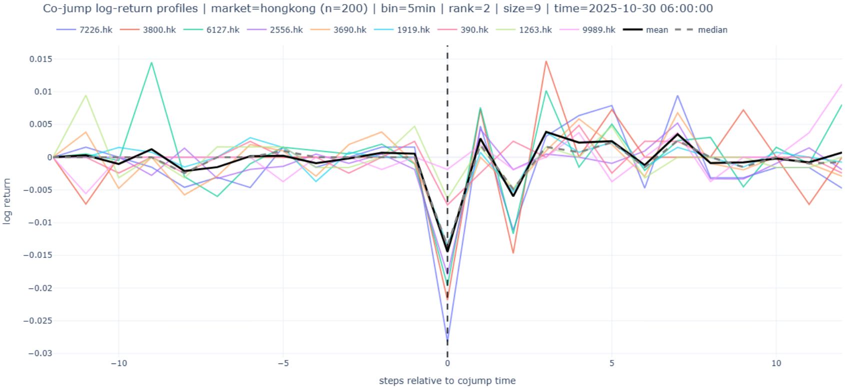
INCONCLUSIVE

Research question 3: robustness across markets (and co-jumps)

- We test multiple markets.
- We connect endo/exo signatures.
- We analyze co-jumps.

Co-jumps: size distribution and alignment

Hypothesis: Co-jump sizes are heavy-tailed.
Plot: Size histogram/CCDF and sign alignment.
What we see: Large co-jumps occur and structure emerges.



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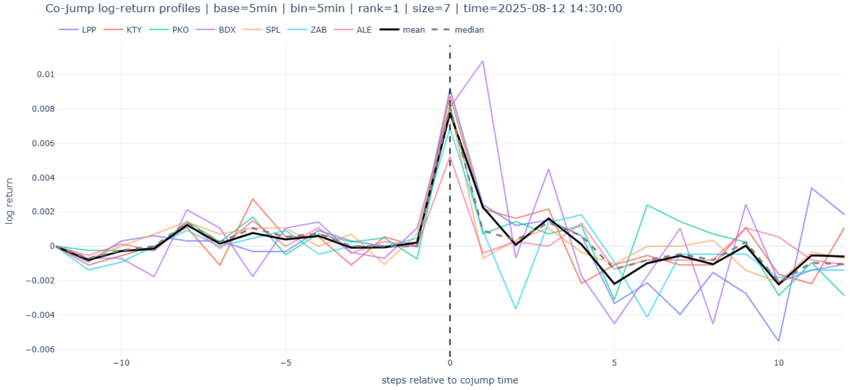
Co-jumps: reflexivity indicator

Hypothesis: Large co-jumps show strong reflexivity.
Plot: Co-jump indicator from aggregated D_1.
What we see: Reflexivity signal strengthens with size.



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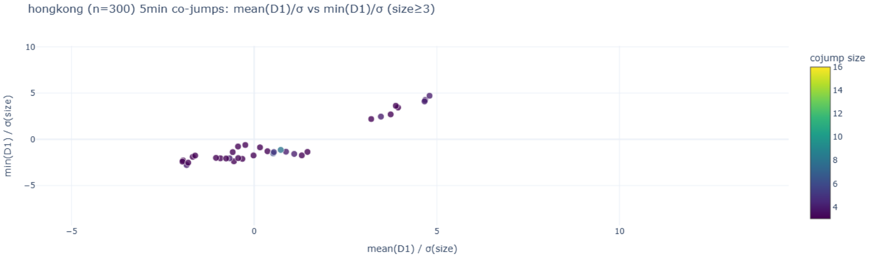
Co-jumps: representative profiles

Hypothesis: Large co-jumps propagate across stocks.
Plot: Representative constituent profiles.
What we see: Synchronized moves cluster around events.



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Takeaways

- We recover jump classes.
- We interpret D_1 reflexivity.
- We see mean-reversion and trend.
- We confirm heavy tails.
- We confirm threshold robustness.
- ✓ We verify our hypotheses.

Questions?